

Jiseok Chae | 채지석

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Academic Positions

Natural Science Research Institute, KAIST

Postdoctoral Researcher

Mar. 2026 – Present

Education

KAIST

Integrated M.S./Ph.D. Program, Mathematical Sciences

Mar. 2021 – Feb. 2026

GPA : 4.24/4.3

Thesis : Two Theoretical Advances in Generative Models: minimax optimization methods with improved convergence and provable benign overfitting

Advisor : Prof. Donghwan Kim

KAIST

Bachelor of Science, Mathematical Sciences

Mar. 2016 – Feb. 2021

Graduated *summa cum laude*, GPA 4.06/4.3

Minor in Computer Science

Professional Interests

- Continuous Optimization for Machine Learning
- Design and Analysis of Machine Learning Algorithms
- Generative Models and Understanding How They Work

Publications

Peer-reviewed.....

Optoelectronic Synaptic Transistors Based on Colloidal CdSe Nanowires for Energy-Efficient Neuromorphic Computing

Woosik Kim*, **Jiseok Chae***, Taesung Park, Hanseok Seo, Hyung Jin Choi, Byung Ku Jung, Seungshin Lim, and Soong Ju Oh

*equal contribution

ACS Applied Materials & Interfaces, 18(5), 2026

Stochastic Extragradient with Flip-Flop Shuffling & Anchoring: Provable Improvements

Jiseok Chae, Chulhee Yun, Donghwan Kim

38th Annual Conference on Neural Information Processing Systems

NeurIPS 2024

Two-timescale Extragradient for Finding Local Minimax Points

Jiseok Chae, Kyuwon Kim, Donghwan Kim

Twelfth International Conference on Learning Representations

ICLR 2024

HEaaN-STAT: a privacy-preserving statistical analysis toolkit for large-scale numerical, ordinal, and categorical data

Younho Lee, Jinyeong Seo, Yujin Nam, **Jiseok Chae**, Jung Hee Cheon

Open Problem: Is There a First-Order Method that Only Converges to Local Minimax Optima?*Jiseok Chae, Kyuwon Kim, Donghwan Kim*

36th Annual Conference on Learning Theory (Open Problems Track)

COLT 2023

Workshops/Preprints**Understanding Schedule-Free Methods in Nonconvex Optimization: Rate Guarantees and Escaping Saddles***Jiseok Chae, Donghwan Kim*

Preprint.

Rethinking Memorization–Generalization Trade–Off in Generative Models*Jiseok Chae*, Kyuwon Kim*, Donghwan Kim*

*equal contribution

3rd Workshop on High-dimensional Learning Dynamics, ICML 2025

Professional Experience**CryptoLab, Inc.**

Seoul

*Research Programmer Internship**Jun. 2019 – Aug. 2019*

- Developed HEaaN.STAT, a statistical data analysis program using the homomorphic encryption scheme HEaaN. Joint work with Korea Credit Bureau

Skills**(Programming) Languages & Software**

- Python (NumPy, PyTorch, ...)
- C++
- MATLAB
- L^AT_EX

(Natural) Languages

- Korean (Native)
- English (Fluent)
- Japanese (Intermediate)

Invited/Contributed Talks**Constructing Convergent Methods for Minimax Optimization Problems**

KIAS AI and Natural Sciences Center Seminar; Korea, 2026

Yonsei University Computational Science and Engineering Seminar; Korea, 2026

Schedule-Free Methods are Rate-Optimal in Nonconvex Optimization

2026 KSIAM Spring Conference; Korea, 2026

Rethinking Memorization–Generalization Trade–Off in Generative Models

2025 Korean Mathematical Society Annual Meeting; Korea, 2025

Stochastic Extragradient With Flip-Flop Shuffling & Anchoring

2025 International Conference on Continuous Optimization; USA, 2025

2024 Korean Mathematical Society Spring Meeting; Korea, 2024

Two-timescale Extragradient for Finding Local Minimax PointsKAIST Stochastic Analysis and Application Research Center *CoLabor* Workshop; Korea, 2023

Selected Honors, Awards, and Grants

Jang Young Sil Fellow Program (Excellence Track)	
KAIST	2026
Outstanding Paper Award	Doctoral dissertation
KAIST	2026
National Academic Excellence Scholarship (Science and Engineering)	
Korea Student Aid Foundation (KOSAF)	2016 – 2020
Dean's List	
College of Natural Sciences, KAIST	Spring 2019
School of Freshman, KAIST	Spring 2016

Teaching Experience

KAIST	
○ Head Teaching Assistant	
- MAS110 Linear Algebra for Data Science	Spring 2022 – Spring 2024
· The founding head TA of the newly established course.	
○ Programming Teaching Assistant	
- MAS109 Introduction to Linear Algebra	Spring 2022 – Spring 2024
· Taught MATLAB; designed course materials and programming assignments.	
○ Teaching Assistant	
- MAS.10002 Calculus II	Fall 2025
- MAS473 Introduction to AI with Mathematics	Fall 2024
- MAS374 Optimization Theory	Fall 2021
- MAS101 Calculus I	Spring 2021 – Fall 2021

Additional Information

○ Technical Research Personnel (전문연구요원)	Mar. 2024 – (Feb. 2027)
○ Regular member of the Korea Cube Culture United (KCCU)	